

Anupam Siwakoti

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Education

Lamar University, MS in Computer Science

Aug 2023 – May 2025

GPA: 4.0 • Coursework: Machine Learning, Data Science, Cloud Computing

Leeds Beckett University, BSc(Hons) in Computing

Sept 2018 – May 2022

GPA: 3.8 • Coursework: Advanced Database Systems, Advanced Software Engineering

Skills

Languages: Python, C++, Java, SQL (MySQL, PostgreSQL)

Machine Learning / Data: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, TensorNetwork (MPS), Oracle APEX

Frameworks / Tools: FastAPI, Django (DRF), Docker, Redis, Kafka, Grafana, GitHub, JIRA

Awards and Certifications

Machine Learning Specialization (Andrew Ng, Stanford) • Google Cloud ML Engineer Professional Certificate

Experience

Software Developer, Center for Midstream Management and Science (CMMS) – Beaumont, TX

June 2025 – Current

- Developed a Physics-Informed ML system for real-time anomaly detection and forecasting, combining Vector Autoregression (VAR) with a patented tensor train fingerprinting approach; validated on a physical testbed for complex fluid flows, achieving 0.5 ms detection and 1–10 ms forecasting speeds.
- Designed a data-driven stochastic model predictive control (SMPC) architecture that leverages Functional PCA and Normalizing Flows to predict joint multi-variable distributions, minimizing fuel consumption in turbocompressor trains.
- Containerized the entire anomaly-detection pipeline (Kafka producers/consumers, C++ forecaster, Python analytics) using Docker Compose to enable reproducible multi-service deployments and cut teammate onboarding time by 50%.
- Deployed Prometheus + InfluxDB monitoring across the streaming stack, exposing custom latency and fault metrics plus Grafana dashboards that reduced anomaly triage from hours to minutes.
- Built full-stack embedded Linux solutions on **Octavo OSD32MP1-BRK** platform, integrating SparkFun Babysitter, Murata Cat-M1/NB-IoT, and I2C-based sensors.
- Programmed **TGS 8410 methane sensors** using STM32CubeIDE and ARM Cortex-M4 firmware, implementing precise heating cycles and calibrated ADC sampling.

Graduate Research Assistant, Lamar University – Beaumont, TX

May 2024 – May 2025

Collaboration with Jefferson Lab, Virginia

[Github](#)

- Built ML pipeline with **Optuna + PyTorch** for HPO of DNNs on GlueX detector data, improving particle-ID accuracy by 25%.
- Developed modular PyTorch MLP framework with schedulers, gradient monitoring, and diagnostics, reducing training/eval cycles by 40%.
- Enhanced high-energy physics analysis with ML-based particle categorization, improving scientific insights.
- Deployed PID models in C++ within HallD software, enabling real-time nuclear physics data analysis.

Software Engineer, Codavatar – Kathmandu, Nepal

Nov 2021 – June 2023

- Backend engineer for Dialaxy, creating secure authorization modules (OSO + Python) and Twilio-powered APIs.
- Built full-stack solution (GraphQL, Pandas, FastAPI, Docker) for developer performance analytics, completing 50% faster than planned.
- Designed REST APIs with Django REST Framework (DRF) for Entegra, enhancing system functionality.
- Developed asynchronous API with FastAPI and secure token management for authentication.
- Built web app with Django + PostgreSQL, responsive frontend (HTML, CSS, JS), and async ops with AJAX.

Projects

Physics-Informed Real-Time Predictive Control System

[Github](#)

Python, PyTorch, Normalizing Flows, Tensor Networks, SCADA, Pandas, NumPy, Apple Silicon (MPS). Developed a stochastic model predictive control (SMPC) architecture for real-time energy optimization in LNG turbocompressor pipelines. Integrated deterministic physical simulations (digital twins) with Conditional Normalizing Flows to accurately forecast multivariate operational uncertainties. Optimized the data processing and machine learning pipeline for edge hardware, reducing probabilistic forecasting and control decisions to milliseconds.

Industrial Time-Series Anomaly Detection & Forecasting Platform

[Github](#)

C++, Python, Docker, Apache Kafka, Grafana, Prometheus, CMake, Git, Pandas, NumPy, Matplotlib. Developed an end-to-end anomaly detection and forecasting system for industrial time-series data using advanced statistical and machine learning techniques. Designed and implemented a modular C++ backend for real-time data processing, integrated with Python for data preprocessing, feature engineering, and result visualization. Leveraged Docker for containerized deployment, Kafka for data streaming, and Grafana/Prometheus for monitoring and visualization.

UK Crime Analytics Data Warehouse & BI Dashboard

[Github](#)

Oracle APEX, PL/SQL, Tableau. Consolidated siloed regional crime databases into a centralized Star Schema Data Mart by developing automated, data-cleansing PL/SQL ETL pipelines. Leveraged this high-integrity repository to build interactive Tableau and Excel dashboards, providing executives with drill-down visualizations of crime hotspots, case resolution times, and officer efficiency to track "Reduce Crime" KPIs.

AI-Powered Real-Time Anomaly Detection

[Github](#)

FastAPI, Docker, Kafka, Redis, WebSockets, PostgreSQL, Nginx, ML, Prometheus, Grafana. Real-time service with Kafka/WebSockets; DDD design; Redis caching, PostgreSQL + Alembic migrations.